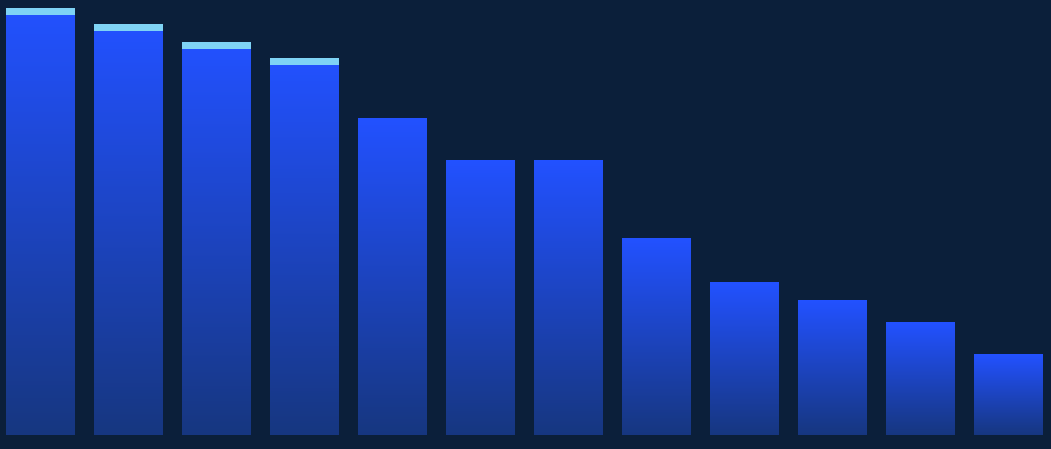
PORTFOLIO CASE STUDY

OCTOBER 2026

What drives house prices in Hanoi?

A regression study of 11,294 house listings: how much location, size, house type and legal status move the asking price



Each bar is a district's asking-price premium over Hoài Đức at equal size, from Hoàn Kiếm (+2.13bn, left) to Hà Đông (+0.40bn).

Tran Bao Phuc

Python · pandas · statsmodels · scikit-learn

VIEW CODE ON GITHUB

Location sets the price level; floor area, house type and legal status adjust it

01 **Location dominates.** At equal size, type and legal status, a house in Hoàn Kiếm, Cầu Giấy, Ba Đình or Đống Đa asks about VND 2bn more than one in suburban Hoài Đức.

02 **Averages mislead.** Cầu Giấy has the higher median price, but Hoàn Kiếm houses are 22% smaller; at equal size Hoàn Kiếm is at least as expensive.

03 **Attributes add up.** Each extra m² adds ~VND 76m and a land-use certificate ~0.9bn; an alley house asks ~1.1bn less than a street-front one.

04 **Just over half explained.** Test R^2 0.55–0.58 and a typical error of ~VND 0.8bn; the rest likely lies in what listings omit, such as alley width and condition.

AT A GLANCE

≈ 2bn
VND inner-city premium vs Hoài Đức

+76m
VND per extra m² of floor area

+0.9bn
VND for a land-use certificate

55–58%
of price variation explained

SO WHAT

Buyers should compare districts at equal size, not by average price; sellers gain ~VND 0.9bn by completing the land-use certificate; and better location data (alley width, distance to the centre) is the clearest way to improve price estimates.

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11,294 listings were cleaned to 9,862 houses before modelling what drives price

DATA

11,294 Hanoi house listings: district, ward, house type, legal status, floors, bedrooms, floor area, asking price (VND bn).

PERIOD

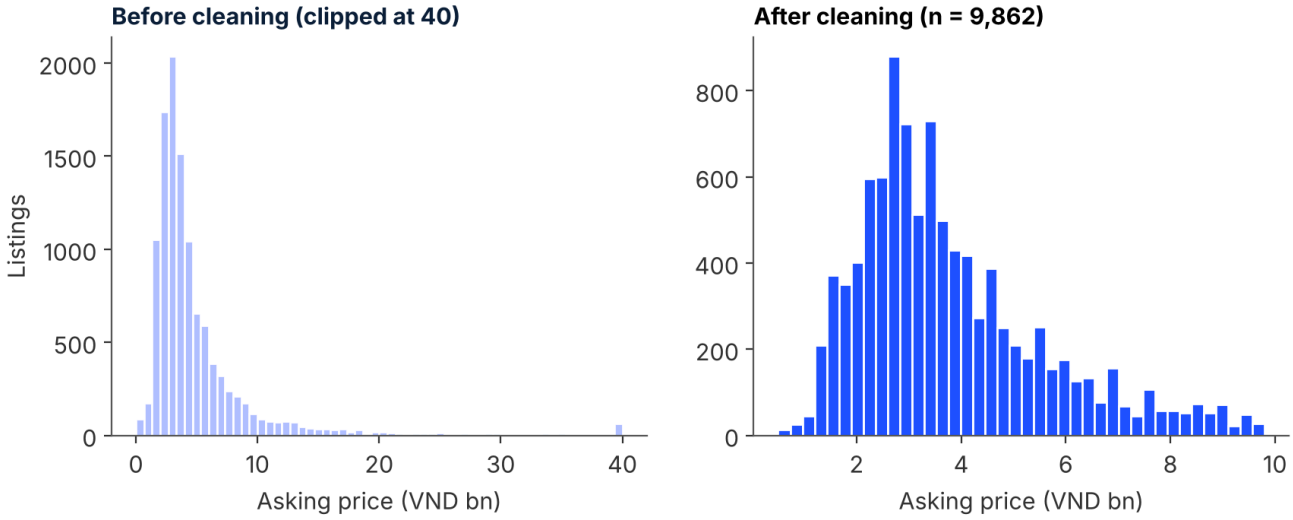
Listings posted July–August 2020; course dataset from Tomorrow Marketers.

METHOD

Multiple linear regression (OLS), 80/20 train–test split, assumption checks, 5-fold cross-validation.

EXHIBIT 1

Cleaning removed implausible prices and left a right-skewed but usable distribution



Source: Hanoi house listings, Jul–Aug 2020; n = 11,294 before and 9,862 after cleaning

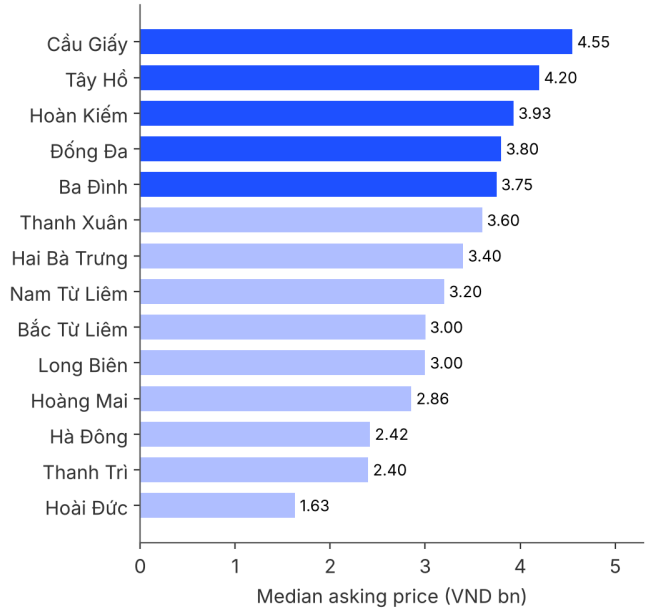
Data issues and fixes

ISSUE FOUND	FIX	EFFECT
Prices from 0 to VND 100bn	IQR rule on price and floor area; floor at VND 0.5bn	−1,432 rows
Length and width typed in mm (“120000 m”)	Dropped; floor area already carries the information	−2 columns
Districts with under 30 listings (e.g. Sóc Sơn: 3)	Grouped as “Other”	8 → 1 group
Text fields such as “4 phòng”, “10 m”	Converted to numbers	3 columns

Inner-city medians are up to 2.8× suburban ones, but medians mix location with house size

EXHIBIT 2

Median asking price by district (≥ 30 listings)



Source: Hanoi house listings, Jul–Aug 2020 (n = 9,862)

2.8×

gap between the median house in **Cầu Giấy (4.55bn)** and in **Hoài Đức (1.63bn)**.

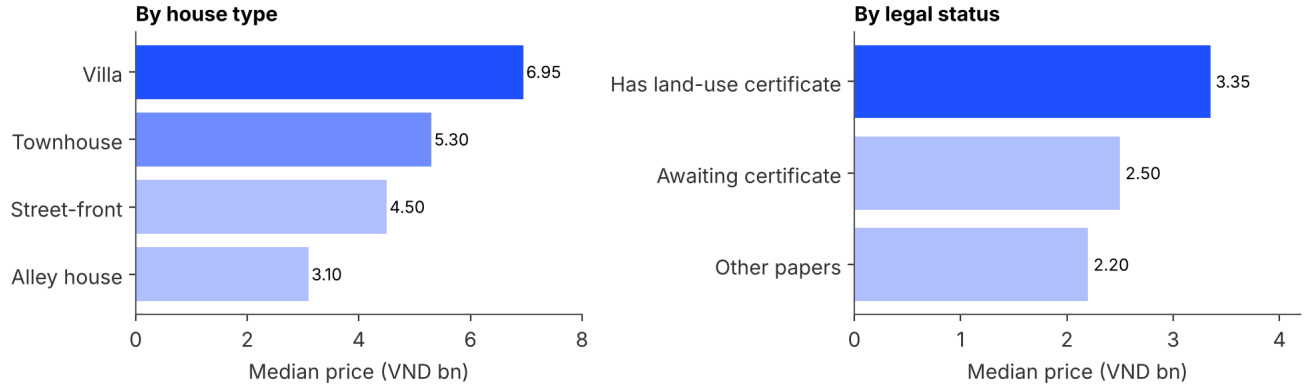
- The top five districts form the inner-city core: Cầu Giấy, Tây Hồ, Hoàn Kiếm, Đống Đa, Ba Đình.
- Suburban Hà Đông and Thanh Trì sit at ~VND 2.4bn.

CAUTION

Hoàn Kiếm houses are 22% smaller than Cầu Giấy's, so its median understates its price level. The regression corrects for this (page 6).

EXHIBIT 3

Villas and certified houses ask the most; alley houses, 78% of listings, the least



Source: Hanoi house listings, Jul–Aug 2020 (n = 9,862). Only 1.3% of listings lack a land-use certificate, but the gap is consistent, so it is kept as a 0/1 flag.

Floor area, bedrooms, district and house type enter the model; length and width are dropped

EXHIBIT 4

Floor area has the strongest link with price



Pearson correlation, n = 9,862

Model set-up

- **Target:** asking price (VND bn)
- **Numeric:** floor area, bedrooms, floors
- **Categorical** (one-hot, drop-first): district, baseline **Hoài Đức**; house type, baseline **villa**
- **Binary:** has land-use certificate
- 21 variables; train 7,889 / test 1,973

CHECK

All variance inflation factors are below 1.4, so the predictors do not duplicate each other.

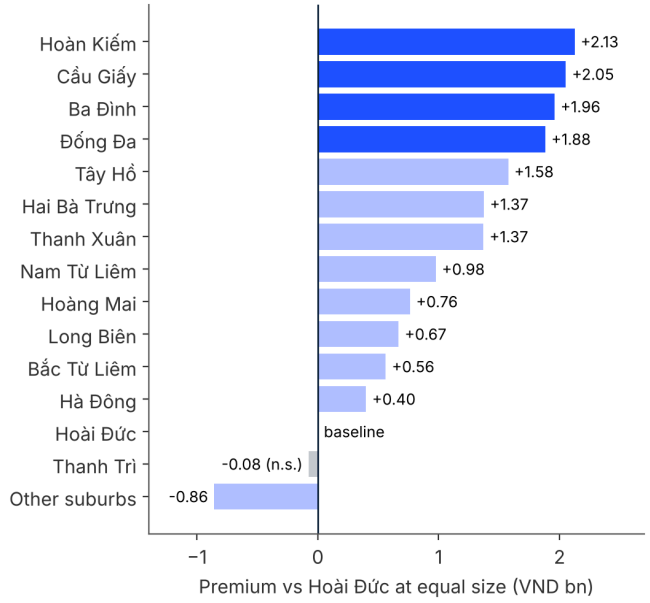
Variable decisions

VARIABLE	DECISION	REASON
Floor area	**Kept**	Strongest numeric link with price (r = 0.59)
Bedrooms	**Kept**	r = 0.42; only 0.38 with floor area, so it adds information
Floors	**Kept**	Weak alone (r = 0.17) but a useful control
District, house type	**Kept**	Largest price gaps in Exhibits 2–3
Length, width	**Dropped**	Unit errors (r ≈ 0.00) and duplicate of floor area
Address, date	**Dropped**	Free text; one-month window
Ward (207 groups)	Model 2 only	Tested separately to avoid over-fitting the base model

At equal size, inner-city districts ask about VND 2bn more than suburban Hoài Đức

EXHIBIT 5

Premium vs Hoài Đức, holding size, type and legal status constant



Source: OLS regression on 7,889 training listings; n.s. = not significant

+2.13bn

for a house in **Hoàn Kiếm** vs an identical one in Hoài Đức, the largest premium in the city.

THE RANKING FLIPS

Cầu Giấy leads on median price (4.55 vs 3.93bn), but its houses are larger (44 vs 34.5 m²). At equal size Hoàn Kiếm is at least as expensive; the 0.08bn gap is within the margin of error.

Property attributes, holding location constant

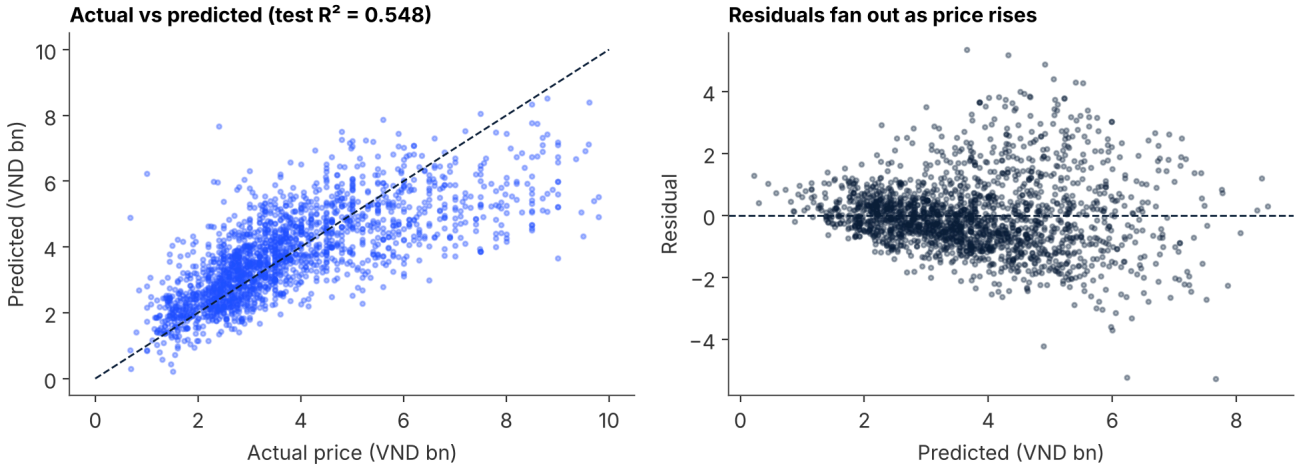
ATTRIBUTE	EFFECT ON ASKING PRICE	READING
Floor area	**+VND 76m** per m²	≈ 2.3bn for 30 m² more
Land-use certificate	**+VND 0.91bn**	the market prices legal risk
Alley vs street-front	**−VND 1.08bn**	same district and size
Each extra bedroom	**+VND 0.22bn**	floor area fixed
Each extra floor	**+VND 0.06bn**	smallest effect

All attribute coefficients are significant at $p < 0.001$. Thanh Trì ($p = 0.62$) and townhouse vs villa ($p = 0.14$) do not differ significantly from their baselines.

The model explains 55–58% of price variation on unseen data but under-prices expensive homes

EXHIBIT 6

Predictions track actual prices up to ~VND 6bn; errors widen above that



Test set, n = 1,973 (Model 1)

Accuracy on unseen data

METRIC	DISTRICT	+ WARD
Variables	21	115
Test R^2	**0.548**	**0.576**
MAE (VND bn)	0.85	0.81
5-fold CV R^2	0.56	0.59

Assumption checks

CHECK	RESULT
Multicollinearity (VIF)	**Passed** · < 1.4
Over-fitting (train–test)	**Passed** · 2–3 pts
Independence (Durbin–Watson)	**Passed** · 1.99
Constant variance (Breusch–Pagan)	**Not met** · p ≈ 0
Normality (Jarque–Bera)	**Not met** · right skew

SO WHAT

The ward model wins in all 5 folds, so the gain from finer location is stable. Use the model for **relative comparisons** between districts and attributes, not to value individual homes above ~VND 7bn, which it under-prices by ~2.5bn on average.

Location should anchor any valuation; finer location data is the next step

FOR BUYERS

Compare districts at equal size. A lower median can simply mean smaller houses, as in Hoàn Kiếm.

FOR SELLERS

Completing the land-use certificate adds ~VND 0.9bn; street-front access adds ~1.1bn.

FOR RESEARCH

District explains much; ward adds 3 points of R^2. Finer location data is the next lever.

Limitations

- Asking prices, not transaction prices; listings contain entry errors.
- 2020 data, before the 2021–2024 price rise; not suitable for valuing homes today.
- Scope: houses up to 75 m² and VND 10bn; apartments and large villas excluded.
- About 42% of price variation remains unexplained.

Next steps

#	ACTION	WHY
1	Add alley width and distance to the centre (geocoding)	Largest unexplained share is location detail
2	Test tree-based models	Capture area × district interactions
3	Robust (HC3) errors or log(price)	Residual variance grows with price
4	Refresh with current listings	2020 prices are out of date

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Code and notebook
github.com/phuctran120116-stack/hanoi-housing-price-regression